

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 17: Forest: Our Lifeline

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	17 — Forest: Our Lifeline
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Watch for second-order food-web ripples that loop back, nutrient-cycle breaks dressed as harmless, and answers that confuse energy flow with nutrient flow or treat any single forest service in isolation. Do not write on this paper — mark answers on your answer sheet.

1. A botanist measures light at four heights in a forest and finds: emergent treetops 100 percent, mid-trunk shrub level 25 percent, knee-high herbs 8 percent, ground litter 2 percent. A newly sprouted seedling there carries enough stored food to grow only a short while before it must photosynthesise on its own. For best survival odds, the seed is most likely adapted to germinate where it can:

- (A) germinate immediately on the 2 percent floor and complete its whole life there in deep shade
- (B) wait dormant on the dim floor and surge upward only when a falling tree opens a bright gap above it
- (C) climb a grown trunk to the 100 percent treetop before it has any leaves of its own
- (D) grow fastest exactly at the 8 percent herb level because partial shade gives the most food

2. A forest web has: grass eaten by hare and by mouse; hare eaten by fox; mouse eaten by owl and by fox; fox eaten by no one. A student claims 'every animal here sits in only one straight chain.' The single fact that disproves this is that:

- (A) the fox draws food from two separate chains, eating both the hare and the mouse, so the chains cross at the fox
- (B) the owl eats two kinds of prey, so it alone links the chains together
- (C) the grass appears twice, which means the producers form the crossing point
- (D) the mouse eats both grass and the hare, joining the two chains in the middle

3. In the same web (grass to hare/mouse; hare to fox; mouse to owl and fox), suppose the fox is the only predator the hare has, while the mouse has two predators. If a hunter removes all the foxes, the hare and the mouse are released from a predator. Which prediction is most defensible?

- (A) both hare and mouse rise by exactly the same amount because both lost the fox
- (B) the mouse rises more steeply than the hare because the owl now hunts it less
- (C) neither changes, because removing a top predator never affects its prey
- (D) hare numbers are likely to climb more sharply than mouse numbers, since the hare lost its only predator while the mouse still has the owl

4. A web reads: leaves to caterpillar to warbler to hawk, and leaves to caterpillar to spider to warbler. The warbler therefore eats both caterpillars and spiders, and the spider eats caterpillars. A wasp wipes out almost all the spiders. The most complete first prediction is that:

- (A) caterpillars, freed of the spider, tend to rise, while the warbler leans more heavily on caterpillars after losing its spider prey
- (B) the warbler starves out at once because the spider was its only food
- (C) the leaves vanish immediately because the spiders had been protecting them
- (D) the hawk increases straight away because spiders had been competing with it for warblers

5. A forest supports about 50000 kg of grass, which feeds enough deer to total 5000 kg, which feed tigers totalling 500 kg. A student says 'we could double the tigers just by protecting them better.' The deepest reason this is hard is that:

- (A) tigers refuse to breed once their numbers reach a fixed limit set by instinct
- (B) only a small fraction of energy passes up each step, so the grass-and-deer base can support only a limited tiger mass no matter how well they are guarded
- (C) protecting tigers makes deer multiply, and crowded deer cannot be eaten
- (D) tigers stop needing food once their population is large enough

6. Two neighbouring plots get the same leaf fall each autumn. In plot X the litter layer stays thin year after year; in plot Y, after a pesticide killed the soil fungi and bacteria, the litter grows into a thick mat that never fully rots. Reading both plots together, the best single conclusion is that:

- (A) plot X has stronger wind that carries every fallen leaf away before it can settle
- (B) in plot X decomposers break litter into humus at about the rate it falls, while in plot Y the loss of decomposers stalls that recycling and locks nutrients in the pile

(C) plot Y's trees stopped dropping leaves once the litter began to build up

(D) the thick litter in plot Y proves its soil is the more fertile of the two

7. A web: clover to grasshopper to frog to snake; clover to rabbit to fox; and snake and fox are both eaten by the same eagle. A flood wipes out almost all the frogs. Tracing the ripples, the most likely chain of effects is that:

(A) the eagle dies out at once because the frog was the base of all its food

(B) snakes lose their frog food and decline, so the eagle leans more on foxes, while grasshoppers, less eaten, tend to rise

(C) the rabbit population collapses immediately because it shared the frog's predators

(D) the clover is wiped out next because frogs had been fertilising it

8. On a hillside, an intact forest soaks rain into the soil so a spring below flows all year. After felling, three things change together: the bare slope erodes, the spring floods then dries, and less moisture rises into the local air. The single underlying change that drives all three at once is that:

(A) with roots and leaf litter gone, rain no longer soaks in slowly but rushes off, carrying soil and leaving little stored water to feed the spring or to transpire back into the air

(B) the missing trees had been drinking the floodwater and breathing it out as the spring's flow

(C) removing the canopy lets sunlight bake the soil into a sponge that holds even more water

(D) the felled wood gives off steam, which is the only reason the air loses moisture

9. A naturalist notes that monkeys, fruit bats and hornbills crowd the sunlit upper branches, while a damp tangle of ferns, fungi and beetles fills the deeply shaded ground. Asked why the most light and the most decomposer life sit in different layers, the best answer is:

(A) the ground actually gets the most light, which is why so many organisms gather there

(B) the high animals carry all the dead matter up so the canopy hosts the most decomposers

(C) light and dead matter are spread evenly, so the two layers are really the same

(D) the upper branches intercept most light for the fruit and leaves the high animals feed on, while the dim floor collects the fallen dead matter that feeds the decomposers below

10. A child argues, 'A forest can never run short of soil nutrients, so we can take all the fallen wood and dead leaves out for firewood with no harm.' The sharpest flaw in this reasoning is that:

- (A) carting away the dead wood and leaves removes the very matter decomposers would have recycled into humus, so over time the soil loses the nutrients it normally gets back
- (B) it is correct, because plants make all their nutrients fresh from sunlight and never need the soil
- (C) removing dead leaves helps, because rotting matter only poisons the soil for new plants
- (D) the soil never loses nutrients anyway, since rain delivers a fresh supply from the sky each day

11. At midnight in a dense forest, the plants are taking in oxygen and releasing carbon dioxide. A student concludes, 'So forests actually make the air worse, not better.' The precise correction is that:

- (A) plants never take in oxygen, so the student must have observed something else
- (B) the night-time release proves forests harm the air, just as the student says
- (C) plants stop breathing entirely in daylight, which exactly cancels the night-time effect
- (D) by day the same plants take in far more carbon dioxide and release far more oxygen through photosynthesis than they use at night, so over a full day the forest adds oxygen overall

12. A region's grassland is left undisturbed for sixty years: first more herbs appear, then shrubs, then scattered young trees, then a closed canopy with rich humus and many animals. A boy says this proves a forest is a fixed thing you either have or do not. The better reading is that:

- (A) the forest was fixed from the moment the first herbs appeared and never really changed
- (B) a forest is dynamic, developing through stages over many years and renewing itself, rather than appearing fully formed all at once
- (C) the whole sequence shows a forest can spring up complete within a single season
- (D) only deliberate planting in that exact order could ever produce such a forest

13. A teacher draws the loop soil to roots to plant tissue to grazing animals to dung and dead bodies to decomposers to soil, then asks which single cut would most directly stop nutrients ever returning to the soil. The best answer is removing the:

- (A) herbivores, since they are the only link that turns plant tissue into humus
- (B) decomposers, since without them dead bodies and dung are never broken down into humus and the nutrients stay locked away from the soil
- (C) rainfall, since rain is what carries nutrients out of the dead matter and into the roots
- (D) sunlight, since the loop cannot turn at all once the light driving it is cut off

14. A web: leaves to aphids to ladybirds to spiders, and leaves to aphids to ants (which farm and protect aphids from ladybirds). A drought kills almost all the ants. The most reasoned first prediction is that:

- (A) without ant protection, ladybirds eat aphids more freely, so aphid numbers tend to fall and spiders gain more ladybird prey
- (B) aphids rise sharply, because losing the ants removed their only food source
- (C) the leaves are stripped bare, because ants had been the leaves' main grazers
- (D) the spiders die out, because the ants had been their chief food all along

15. A forester explains why deep forest soil stays rich while a cleared, ploughed field beside it grows poorer each year. The best combined reason is that in the forest:

- (A) the canopy keeps out all rain, and the constant dryness alone is what makes soil fertile
- (B) animals churn the soil with their feet, which is the sole source of its richness
- (C) the deep shade by itself enriches the soil, since darkness creates nutrients
- (D) fallen leaves and dead matter are steadily turned into humus by decomposers, while living roots and litter hold that fertile soil in place against rain

16. A tree in this forest is pollinated only by one moth, and its seeds are carried off and planted only by one bird that eats its fruit. Hunters remove that bird while the moth survives. Over several years the most likely outcome for this tree species is that:

- (A) fruit is still set thanks to the moth, but with no bird to disperse the seeds, few new saplings start away from the parent, so the species spreads and renews poorly
- (B) the tree is unaffected, because pollination and seed dispersal are not needed for it to spread
- (C) the tree spreads faster, because all its seeds now fall and grow right beneath the parent
- (D) the tree dies out at once, because losing the bird also stops the moth from pollinating it

17. Comparing services, a planner notes a forest gives timber and honey (products) and also oxygen, soil-holding, water recharge, climate moderation and wildlife shelter (services). A logger argues the forest's whole worth is just the timber. The best counter is that:

- (A) the standing forest delivers many life-supporting services at once that a stack of cut timber cannot, so its value far exceeds the wood alone
- (B) the logger is right, because services like clean air and water have no real value to people
- (C) the products and services are the same thing, so counting timber already counts all of them
- (D) the forest's only genuine value is as a home for wild animals, not for any human use

18. After a hill catchment is deforested, downstream villagers report, in the same few years: muddier flood water in the monsoon, a dry stream in summer, hotter afternoons, and poorer crops on the slope. Linking these, the best single explanation is that losing the forest:

- (A) made the rain itself heavier and dirtier, which alone explains every change reported
- (B) caused only the flooding, since erosion, summer drought and heat have no link to forests
- (C) let rain run off fast (flash floods and lost summer flow), stripped the topsoil (muddy water and poorer crops) and cut transpiration and shade (hotter air), all flowing from one loss of tree cover
- (D) improved the soil and climate, so the villagers must have misread the changes

19. A web: phytoplankton to small fish to kingfisher in a forest pond, and grass to deer to leopard on the bank, with one decomposer community on the floor that breaks down whatever dies in either place. A spill kills most of the small fish. Reading the shared decomposer link, the most accurate statement is that:

- (A) the deer and leopard collapse at once, because they shared the small fish as food
- (B) the pond and bank chains are completely separate, so nothing connects them in the forest
- (C) the kingfisher loses food and tends to decline while ungrazed phytoplankton may build up, and the floor decomposers still recycle the dead from both the pond and the bank into one soil
- (D) each chain keeps its own private decomposers, so the pond's dead never reach the bank's soil

20. A forest manager must pick the action that best keeps a forest's many services running for the long term. The best choice is to:

- (A) remove the forest's decomposers so nothing rots and the forest is kept clean and tidy
- (B) clear the canopy trees so far more sunlight reaches and feeds the forest floor
- (C) convert the forest to a single fast-growing crop so the land yields more each season
- (D) protect the standing forest and replant trees on land where forest has been lost, so the layered community and its services can persist and recover

21. A tree relies on bats to pollinate its night-opening flowers and on a beetle whose grubs eat only that tree's rotting fallen fruit, returning the nutrients to soil. A pesticide kills the beetles while the bats remain. The most likely consequence is that:

- (A) no flowers are pollinated, because losing the beetle also stops the bats from working
- (B) the tree grows faster, because the fallen fruit now stays whole and feeds the roots directly
- (C) the tree is wholly unaffected, since neither pollination nor decay matters to a tree
- (D) flowers are still pollinated by bats and fruit still forms, but fallen fruit now rots more slowly without the beetle grubs, so nutrient return near the tree is delayed

22. In the chain grass to grasshopper to frog to snake to hawk, energy from the Sun is captured by the grass and passed up step by step, with much lost as heat at each transfer.

Which animal's body, on average, traces back to the smallest share of the Sun's original captured energy?

- (A) the grasshopper, because being the smallest-bodied animal it must rely on the least energy
- (B) the frog, because animals in the middle of a chain always receive the least energy of all
- (C) the hawk, because energy is lost at every transfer and the hawk sits at the top, furthest from the producer
- (D) the grass, because producers are given the smallest energy share before any animal feeds

23. A naturalist draws arrows for two things in a forest: the path of energy and the path of nutrients. A student insists 'both must point the same way, since both move through the same food chain.' The key correction is that:

- (A) energy flows one way from the Sun up through producers and consumers and is lost as heat, while nutrients cycle round, returning through decomposers to the soil to be used again
- (B) both energy and nutrients return to the Sun in the end, so the arrows do point the same way
- (C) neither energy nor nutrients moves at all once they enter a plant, so no arrows are needed
- (D) nutrients flow one way and are lost, while energy is the thing that cycles back to the soil

24. A web: leaves to caterpillar to thrush, leaves to weevil to thrush, and thrush to hawk. The thrush eats both caterpillars and weevils. A late frost wipes out almost all the weevils. The most defensible first prediction is that:

- (A) the thrush starves out completely, because the weevil was its only food
- (B) the leaves are stripped bare at once, because the weevils had been their main grazers and caterpillars cannot eat them
- (C) the thrush leans more heavily on caterpillars, so caterpillar numbers may be pressed down even though the leaves lost one of their two grazers
- (D) the hawk declines straight away, because the weevils were the hawk's chief food

25. A boy claims, 'A burnt forest patch is no loss, since we can grow a brand-new full forest there in a few months.' The strongest correction draws on the forest being a built-up system, namely that:

- (A) he is right, because scattering seeds yields a complete grown forest within a single season
- (B) the patch will never grow back at all, so the time it takes is beside the point

(C) a forest re-establishes slowly over many years as herbs, then shrubs, then trees return and the soil and animal community rebuild, so it cannot reappear complete in months

(D) the forest will regrow in months, but only if a top predator is released there first

26. A drought destroys almost all the producers in a forest, sparing the animals at first. Within weeks animals at every level are in trouble. The most far-reaching reason is that:

(A) only the plant-eaters are hit, because the meat-eaters get their energy from prey, not from plants

(B) every animal, herbivore and carnivore alike, depends in the end on the producers for energy, so losing the producers cuts the energy base of the whole web

(C) only the top predator is affected, because it alone sits at the end of every chain

(D) nothing serious happens, because the animals can make their own food once the plants are gone

27. A teacher contrasts a season-long wheat field with a centuries-old forest. The single feature that most fundamentally makes the forest a forest and the field not is that the forest is:

(A) simply larger than the field, since area is the only real difference between them

(B) guaranteed to hold at least one large predator, which a field can never have

(C) a many-species, many-layered community that persists and renews itself over many years, not a single crop grown and cleared in one season

(D) free of all insects and animals, while a field is full of them

28. On a dense forest floor a fern with broad, thin leaves thrives in deep shade, while a treetop grass-like plant with narrow leaves grows in full sun. Asked which adaptation helps the fern, the best answer is that broad thin leaves let the shade plant:

(A) catch and use the small amount of filtered light that reaches the dim floor, so it can still make enough food to survive

(B) store extra water for the dry canopy, since floor plants face the harshest sun

(C) make food without any light at all, drawing energy straight from the humus instead

(D) shed the excess sunlight that floods the forest floor through the open canopy

29. A student writes: 'Forests help the climate, so cutting them must change only the temperature, nothing else.' Synthesising the chapter, the best correction is that losing a forest tends to change, together:

(A) air temperature and moisture, rainfall reliability, soil stability, flood and drought patterns, and the air's carbon dioxide level, because the forest influences all of these at once

(B) only the local temperature, since a forest's sole climate role is to provide shade

(C) only the carbon dioxide level, since gases are the one thing forests truly affect

(D) nothing measurable at all, because a single forest is far too small to matter

30. In the chain decaying leaf to fungus to beetle to shrew, a student labels the fungus a producer 'because it comes first.' The precise correction is that the fungus is:

- (A) a producer after all, since whatever starts a chain must be making its own food
- (B) a herbivore, because it feeds on plant material in the form of the dead leaf
- (C) a carnivore, because the beetle that eats it counts the fungus as living prey
- (D) a decomposer, getting its energy by breaking down the dead leaf rather than by capturing sunlight to make its own food

31. A web: grass to zebra to lion, grass to zebra to hyena, and lions and hyenas scavenge each other's leftover kills. A long drought destroys nearly all the grass. The most far-reaching effect, and its reason, is that:

- (A) only the zebra is hit, because lions and hyenas eat meat and so do not need the grass
- (B) only the lion is hit, because it alone sits at the very top of the web
- (C) every animal in the web suffers, because all of them depend, directly through the zebra or through scavenged kills, on the grass as the producer
- (D) nothing is affected, because lions and hyenas can simply make their own food when grass fails

32. A forest's living trees take carbon dioxide from the air and build it into wood, leaves and, through fallen litter, into the soil's humus. A logging company burns a large tract. The most accurate combined effect on the air is that:

- (A) oxygen rises, because with the trees burnt nothing is left in the area to use up the oxygen
- (B) the carbon stored in the wood and litter is released as carbon dioxide, and with the trees gone less carbon dioxide is taken back in, so the gas rises on both counts
- (C) nitrogen rises, because burning the wood releases the nitrogen the trees had stored
- (D) the air is unaffected, since the carbon the trees held simply vanishes when they burn

33. A web: seeds to mouse to owl, seeds to sparrow to owl, and a fox that eats both mice and sparrows. The owl and fox therefore share two prey. A disease halves the sparrows only. Tracing two steps, the most reasoned prediction is that:

- (A) the seeds vanish at once, because sparrows had been the seeds' only consumer
- (B) owl and fox both lean more on mice, so mice are pressed harder and seeds eaten by mice may be spared a little less than before
- (C) the owl and fox both die out, because the sparrow was the sole food of each
- (D) the mice die out too, because losing sparrows removes the mice's food supply

34. A naturalist points out that even a fallen, decaying log is busy with life and is part of the forest's working. The deepest reason a forest can hold soil, recycle nutrients, refresh air, store water and shelter wildlife all at once is that it is:

- (A) simply a very large stock of timber that happens to provide a few extras while it stands
- (B) a chance collection of unrelated trees growing near one another with no working links
- (C) a living, interconnected system whose plants, animals, soil and micro-organisms each support the others, so many services emerge together
- (D) a man-made arrangement that people must rebuild from scratch each year to keep it working

35. Year after year a forest's plants pull nutrients out of the soil to grow, yet the soil does not run dry of them. A child wonders how. The best account is that:

- (A) the plants quietly add brand-new nutrients to the soil out of the sunlight they absorb
- (B) the nutrients never truly leave the soil, since plants only borrow them for a moment
- (C) decomposers keep returning nutrients to the soil from fallen leaves, dung and dead bodies, roughly replacing what the living plants take up
- (D) rain delivers a fresh batch of soil nutrients down from the sky every single day

36. Sunlight that filters through gaps in the canopy reaches the understorey and floor only weakly. Two students argue over whether this dim light matters. The correct view is that the filtered light:

- (A) is harmful to the lower plants and should ideally be blocked out completely
- (B) only warms the floor animals and plays no part in feeding any of the lower plants
- (C) frees the lower plants from needing to take up water or soil nutrients at all
- (D) is just enough for shade-adapted understorey and floor plants to make their own food and survive beneath the canopy

37. A web: clover to beetle to lizard, clover to grasshopper to lizard, and lizard to snake. The lizard eats both beetles and grasshoppers. A pesticide wipes out the beetles. Reading two steps ahead, the best prediction is that:

- (A) the snake dies out at once, because the beetle was the base of all its food
- (B) the lizard relies more on grasshoppers, so grasshopper numbers may fall and the snake, still fed by lizards, is only mildly affected
- (C) the clover is consumed entirely, because the beetles had been holding the grasshoppers in check
- (D) the lizard population doubles, because losing the beetles leaves more clover for it to eat

38. A forest in a wet region returns water vapour to the air through countless leaves and shades the ground from the Sun. Removing it changes the local air. The combined mechanism by which the forest had helped bring rain and keep summers mild is that:

- (A) the trees physically dragged rain clouds down from the sky using their tall trunks
- (B) the trees burned slightly in the heat and the smoke seeded the clouds to make rain

(C) the trees soaked up every bit of the Sun's heat so that the land could never warm at all

(D) transpiration from its leaves added moisture to the air while its shade and cover kept the ground cooler, together favouring rainfall and milder temperatures

39. A bare, ploughed hillside loses much more topsoil in a downpour than the forested slope beside it. Comparing the two slopes, the chain of reasons that best explains the difference is that the forest's:

(A) roots bind the soil and its leaf litter cushions and slows the rain so water soaks in, whereas the bare slope has nothing to grip the soil and rain rushes off carrying it away

(B) trees had been drinking the rainwater, and now that they are gone the soil simply floats off

(C) canopy had baked the soil dry and hard, and only such hard soil ever resists erosion

(D) shade had cooled the rain, and only cool rain is gentle enough to leave soil in place

40. A web: leaves to caterpillar to warbler, leaves to caterpillar to shrew (shrews also eat warbler eggs), warbler and shrew to owl. The shrew thus both eats caterpillars and preys on warblers' eggs. A trap removes all the shrews. The most complete first prediction is that:

(A) the owl starves out, because the shrew was the owl's only food

(B) warblers, no longer losing eggs to shrews, may increase, while caterpillars face one fewer predator, and the owl leans more on warblers

(C) the leaves are devoured entirely, because shrews had been the leaves' main grazers

(D) the caterpillars vanish at once, because they depended on the shrews for survival

41. A child says protecting one big forest will save 'only the trees in it.' Explaining biodiversity, the best correction is that protecting the forest also protects many other species because:

(A) a forest really contains only one kind of organism, so protecting it protects just that one

(B) the forest's variety of life actually comes from the soil minerals rather than the forest itself

(C) its many layers and varied habitats supply homes and food to a great variety of animals, plants and micro-organisms that depend on the forest to live

(D) the animals could survive equally well on bare land, so saving them does not need the forest

42. A web: grass to deer to leopard, grass to hare to leopard, and grass to insect to bird to leopard, so the leopard draws from three chains. A disease halves the deer alone. The most reasoned prediction is that:

(A) the leopard starves out completely, because the deer was its only source of food

(B) the grass is wiped out, because deer had been the only thing keeping the hares in check

- (C) the hares and birds also halve at once, because they shared the deer as a food source
- (D) the leopard shifts more onto hares and birds, so those prey may be pressed harder, while grass once eaten by deer is grazed a little less

43. An island forest is found to recycle nearly all its nutrients on site, taking very little fresh mineral input from outside. A scientist calls it 'self-sustaining.' The best justification is that:

- (A) its plants create entirely new nutrients from sunlight, so no outside supply is ever required
- (B) rain steadily imports all the nutrients it needs, so the forest never has to recycle any
- (C) its animals carry nutrients in from the sea each day, replacing whatever the plants take up
- (D) its decomposers continually return the nutrients in dead matter to the soil, so the same nutrients are reused again and again within the forest

44. A forest captures carbon, holds soil, recharges groundwater, refreshes air and shelters wildlife, and a planner asks where the forest stores most of the carbon it removes from the air. The best answer is:

- (A) in the open air packed between the trees, held there tightly by the canopy above
- (B) in its wood, leaves and soil humus, where the carbon is built into plant tissue and stored as the forest grows
- (C) in the fast-flowing streams that run through the forest, dissolved in the water
- (D) in the fur and feathers of the many animals that live within the forest

45. After a forest fire clears a patch, a student expects either bare rock forever or a full forest by next morning. Observing the patch for several years would most likely show:

- (A) herbs colonise first, then shrubs, then young trees, with soil and animal life rebuilding gradually, so a layered forest returns slowly rather than instantly or never
- (B) the patch stays bare rock indefinitely, because a burnt forest can never recover
- (C) a complete grown forest stands there by the following morning, fully layered overnight
- (D) only grass grows and the patch becomes permanent grassland that no trees can ever enter

46. A web: nectar plants pollinated by bees, bees eaten by birds, birds eaten by snakes. A pesticide kills almost all the bees. Tracing both up and down the chain, the most complete prediction is that:

- (A) only the birds are affected, since pollination and seed-set have nothing to do with the bees
- (B) birds lose food and tend to decline, snakes feel the loss one step later, and the nectar plants are pollinated less, so they set fewer seeds and fruit

(C) the nectar plants grow faster, because with no bees taking nectar they keep more energy for seeds

(D) the snakes increase, because fewer bees means cleaner flowers and healthier prey for them

47. On a slope, an intact forest keeps the water table high and a well below it full all year. After clearing, the well drops and runs dry in summer. The best explanation linking forest to groundwater is that, before clearing, the forest:

(A) turned the rainwater into oxygen, which then condensed underground to fill the well

(B) sealed the soil so tightly that water was forced sideways into the well from the surface

(C) sent all its rainwater straight to the sea, which somehow kept the well topped up

(D) let rain soak slowly through litter and root channels into the ground, recharging the water table that fed the well, whereas bare land sheds rain too fast to recharge it

48. A keystone predator such as the tiger keeps deer numbers in check. Poachers remove all the tigers. Following the ripples through several steps, the most likely far-reaching outcome is that:

(A) more deer means more dung, which instantly turns the whole patch into rich new forest

(B) nothing changes, because the plants and animals in a forest do not affect one another

(C) deer multiply, overgraze the young shrubs and saplings, and the forest struggles to renew itself, showing how one removed species disturbs many others

(D) the surplus deer begin hunting and eating tigers, balancing their own numbers again

49. A naturalist insists that even though the forest floor is the darkest layer, it teems with more crawling and growing life than the brilliant canopy. Resolving this apparent paradox, the best explanation is that:

(A) it is impossible, since the layer with the least light must always carry the fewest living things

(B) the floor must really be the brightest layer after all, because it holds the most life

(C) the floor organisms glow with their own light, so they need none from the Sun above

(D) most light is captured by the layers above, but the floor's piled-up dead matter and damp shelter feed and house vast numbers of decomposers and ground organisms

50. A web: algae to tadpole to fish, leaf litter to insect larva to fish, and fish to heron, so the fish draws from a plant chain and a detritus chain. A pollutant kills the algae only. The most reasoned prediction is that:

- (A) the heron dies out at once, because the algae was the base of all its food
- (B) the fish is unaffected in every way, since algae played no part in its food supply
- (C) tadpoles lose food and decline, but the fish can lean on the litter-fed larvae, so the fish and heron are squeezed rather than wiped out
- (D) the leaf litter disappears too, because the algae had been producing it

51. A teacher labels the loop soil to roots to plant tissue to animals to dung and dead bodies to decomposers to soil a 'cycle' and the line grass to deer to tiger a 'chain.' Asked why 'cycle' fits the first but not the second, the best answer is that:

- (A) in the loop the same nutrients return to the soil to be taken up again and again, while a food chain runs one way and energy is lost rather than returned
- (B) the loop's nutrients travel in a perfect circle of exactly equal size each year, unlike the chain
- (C) a chain may have at most three organisms, whereas a cycle is allowed to have more
- (D) the loop destroys its nutrients at the end and makes a brand-new set, while the chain reuses one set

52. Two villages sit below identical hills; one hill keeps its forest, the other is cleared. Over a decade the forested village enjoys a steady year-round stream, mild summers and fertile fields, while the cleared one suffers floods, dry summers, harsh heat and thin soil. Pulling these together, the best single conclusion is that:

- (A) the difference is pure chance, since a forest cannot influence streams, climate and soil together
- (B) only the flooding differs, because climate and soil fertility have no link to forest cover
- (C) the standing forest simultaneously steadied the water flow, moistened and cooled the air and held the fertile soil, services the cleared hill lost all at once
- (D) the cleared hill is actually better off, so the villagers must have confused the two outcomes

53. A child collects 'forest products' for a project and lists honey, gum, medicinal herbs, timber, plastic and petrol. Asked which entries are truly products obtained from the living forest, the correct group is:

- (A) plastic, petrol and timber, since these are the items people most often buy
- (B) gum, honey and plastic, since all three are sticky or natural-seeming substances
- (C) honey, gum, medicinal herbs and timber, since all come from forest plants and animals
- (D) petrol, medicines and timber, since each is dug or cut from forest land

54. A web: grass to grasshopper to frog to snake to eagle. A farmer sprays a poison that kills the snakes. Tracing the ripple two steps in each direction, the most complete prediction is that:

- (A) the eagle loses snake prey and tends to decline, while frogs, less eaten, rise and press grasshoppers down, easing grazing on the grass
- (B) the grass vanishes at once, because the snakes had been protecting it from the grasshoppers
- (C) the eagle increases straight away, because snakes had been competing with it for frogs
- (D) nothing changes anywhere, because removing a mid-chain animal never affects the rest

55. Asked for the single statement that best captures why a forest is called 'our lifeline,' a student must pick the most complete one. The best choice is:

- (A) it gives us only the timber and firewood that can be cut from it and sold
- (B) it matters only to the wild animals that live in it and not to people at all
- (C) it is just a large group of trees standing together with no wider role to play
- (D) it supplies oxygen, recycles nutrients, holds soil and water, helps balance the climate and shelters countless species, all at the same time

56. Two students name parts of the same forest. Meera points to the leafy roof formed where the branches of many tall trees meet and interlock overhead. Vikram points to the branchy crown of one single tall tree. The terms they are naming, in that order, are the:

- (A) crown of the whole forest, and the canopy of the single tree
- (B) understorey, and the canopy, since both lie below the treetops
- (C) canopy, made by the meeting crowns of many trees, and the crown of one individual tree
- (D) emergent layer, and the forest floor, which face each other top and bottom

57. A web: grass to rabbit to fox, grass to vole to fox and to owl, and the fox and owl share the vole. A disease removes the foxes entirely. Tracing two steps, the most reasoned prediction is that:

- (A) the grass vanishes at once, because foxes had been the only thing keeping the rabbits grazing it
- (B) rabbits, losing their only predator, rise sharply, while the owl now has more voles to itself as the fox no longer competes for them
- (C) the owl declines, because the fox had been the owl's main source of food
- (D) the voles die out, because the foxes used to supply them with food

58. A scientist observes a patch for a decade before judging it a true forest. The single strongest piece of evidence for that judgement would be:

- (A) a very large count of tall trees, all of one single species, packed closely together
- (B) the presence somewhere in the patch of at least one large predator such as a tiger

(C) a thick, even carpet of green grass spread across open ground

(D) many different species of plants, animals and micro-organisms living in interlinked layers and renewing themselves over the whole ten years

59. A pairing quiz lists forest roles against organisms. The pairing that is correct is:

(A) decomposer paired with a mushroom, a fungus that breaks down dead matter into humus

(B) producer paired with a tiger, because the tiger sits at the top of the forest

(C) carnivore paired with grass, because grass feeds the animals beneath it

(D) herbivore paired with a soil fungus, because the fungus lives on plant remains

60. A forest patch loses both its top predator and its decomposers to two separate disasters in the same year. Predicting the combined result, the most complete answer is that:

(A) the forest improves, because fewer predators mean more animals and no rotting to spread disease

(B) the herbivores, unchecked by the predator, overgraze the plants, while dead matter piles up unrecycled and the soil grows poorer, so the forest is hit from two directions at once

(C) only the predator's loss matters, since decomposers have no real effect on a forest

(D) nothing changes, because the two losses cancel each other out exactly

Solutions — Science (Class 7), Chapter 17: Forest: Our Lifeline — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	D	31	C	41	C	51	A
2	A	12	B	22	C	32	B	42	D	52	C
3	D	13	B	23	A	33	B	43	D	53	C
4	A	14	A	24	C	34	C	44	B	54	A
5	B	15	D	25	C	35	C	45	A	55	D
6	B	16	A	26	B	36	D	46	B	56	C
7	B	17	A	27	C	37	B	47	D	57	B
8	A	18	C	28	A	38	D	48	C	58	D
9	D	19	C	29	A	39	A	49	D	59	A
10	A	20	D	30	D	40	B	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) A seedling with limited stored food cannot make enough food in 2 percent light, and it cannot climb to the top without leaves; the herb-level claim wrongly treats dim light as ideal. Many forest-floor seeds stay dormant and shoot up only when a treefall gap floods the spot with light, matching the real light gradient.

2. (A) The owl here eats only the mouse, and grass appearing as the shared base does not make a crossing among consumers; the mouse eats only grass, not the hare. The fox feeding on both the hare-chain and the mouse-chain is what knits the two chains into a web.

3. (D) Losing the fox frees both prey from one predator, but the hare had only the fox while the mouse still faces the owl that keeps it partly in check, so equal rises or a steeper mouse rise are unlikely; the claim that predator loss has no effect contradicts how webs respond.

4. (A) The spider was only one of the warbler's foods, so the warbler does not starve; leaves are eaten by caterpillars, not protected by spiders; spiders did not hunt warblers, so the hawk gains no direct food. Removing the spider lifts predation on caterpillars (they rise) and pushes the warbler to rely more on caterpillars.

5. (B) Tigers do breed, deer being many does not stop them being prey, and tigers always need food; the real ceiling is that energy shrinks sharply at each trophic step, so the producer-herbivore base fixes how much predator mass the forest can carry.

6. (B) Wind removal and trees ceasing leaf-fall are invented causes, and a piling, un-recycled litter mat signals nutrients trapped rather than richer soil. The contrast shows decomposers normally match litter breakdown to leaf fall (plot X); removing them stalls the cycle (plot Y).

7. (B) The eagle eats snakes and foxes, not frogs directly, so it does not starve; rabbits sit in the other chain and did not share the frog's predators; frogs do not fertilise clover. Fewer frogs starve the snakes (decline), shift the eagle toward foxes, and release grasshoppers to rise.

8. (A) Trees do not drink floods to feed springs, baked soil holds less not more water, and cut-wood steam is trivial. One change, loss of the root-and-litter sponge, simultaneously speeds runoff (erosion and flash floods), empties the slow store that fed the spring, and cuts the transpiration that moistened the air.

9. (D) The floor is the darkest layer, not the brightest; dead matter collects on the ground, not in the canopy; and the layers are clearly not uniform. Light is captured high (feeding canopy animals) while fallen dead matter accumulates low (feeding floor decomposers), placing the two riches in different layers.

10. (A) Plants do not make mineral nutrients from sunlight, decomposing litter enriches rather than poisons soil, and rain does not resupply nutrients daily. The forest stays fertile only because decomposers return litter and dead wood to humus; hauling that matter away breaks the loop and drains the soil.

11. (D) Plants do respire at night, so the observation is real, not a mistake, and they do not stop respiring by day. Because daytime photosynthesis takes up much more carbon dioxide and gives out much more oxygen than night respiration consumes, the net effect over a day is to enrich the air with oxygen.

12. (B) The sequence took sixty years, not one season, and it unfolded on its own without a planting plan, so it was neither fixed from the start nor instantly complete. The gradual, self-driven progression from herbs to a layered community shows the forest is a dynamic, self-renewing system.

13. (B) Herbivores are not the humus-making link, rain does not move nutrients into roots from dead matter, and sunlight powers photosynthesis but is not the return step to soil. Only the decomposers convert dung and dead bodies back into humus, so cutting them severs the nutrients' road back to the soil.

14. (A) Ants were protectors of aphids, not their food, so losing ants does not feed-starve aphids upward; ants ate aphid honeydew, not leaves; and spiders ate ladybirds, not ants. Removing the protecting ants exposes aphids to ladybirds, so aphids fall and the better-fed ladybirds offer spiders more prey.

15. (D) A rainless soil is not what makes it fertile, hooves alone do not enrich soil, and darkness does not create nutrients. Forest soil stays rich because decomposers continually add humus while roots and litter protect it from being washed away, two effects the bare field loses at once.

16. (A) Pollination and dispersal both matter, and losing the bird does not stop the surviving moth from pollinating, so neither 'unaffected' nor 'dies at once' fits; seeds dropped only under the parent compete in its shade and rarely spread the species. Without its disperser the tree sets fruit but cannot scatter offspring, so it renews poorly, showing how interdependent the partners are.

17. (A) Clean air and water clearly do have value to people, products and services are not the same (cut timber gives no oxygen or water recharge), and limiting the worth to wildlife ignores human benefit. The living forest supplies many simultaneous services that felled timber cannot, so its value is far more than the wood.

18. (C) Rain did not become intrinsically heavier or dirtier, the listed effects are all forest-linked rather than unrelated, and deforestation worsens rather than improves soil and climate. One cause, the lost tree-and-litter cover, simultaneously speeds runoff, exposes soil to erosion, and removes the transpiration and shade that cooled the air.

19. (C) Deer and leopard never ate the fish, so they do not collapse; the shared decomposer link shows the chains are not wholly separate and do not keep private recyclers. Losing the fish starves the kingfisher (decline) and frees the phytoplankton, while the single decomposer community keeps recycling the dead of both chains into one soil.

20. (D) Removing decomposers would halt the nutrient recycling the forest depends on, clearing the canopy exposes and harms the lower layers, and a single-crop conversion destroys the many-layered community. Protecting the existing forest and replanting lost areas keeps the whole system, and its services, intact and renewing.

21. (D) Bats survive and keep pollinating, so flowering continues and the beetle loss does not disable the bats; whole un-decayed fruit does not feed roots directly, and both

pollination and decay do matter. Without the beetle grubs the fallen fruit breaks down more slowly, delaying the nutrients' return to the soil near the tree.

22. (C) Body size does not set the energy share, the middle of a chain is not the minimum, and the grass as producer captures the most, not the least. Since a fraction is lost at each step, the organism furthest up the chain, the hawk, depends on the smallest remaining share of the Sun's energy.

23. (A) Energy does not return to the Sun and is steadily lost as heat, both clearly do move through the forest, and it is energy, not nutrients, that flows one way. Energy makes a one-way trip and is lost, whereas nutrients are recycled back to the soil by decomposers, so the two arrows differ.

24. (C) The thrush ate caterpillars too, so it does not starve; caterpillars still graze the leaves, so they are not stripped of grazers; and hawks ate thrushes, not weevils. Losing weevils pushes the thrush onto caterpillars more, which can press caterpillar numbers down even as the leaves keep one grazer.

25. (C) Sowing seeds does not yield a grown forest in a season, a cleared patch generally does regrow given time, and no predator is needed to start regrowth. A forest is a slowly assembled, layered community whose soil and species rebuild over many years, so it cannot reappear complete within months.

26. (B) Meat-eaters trace their energy back through prey to the plants, so they are not exempt, and the effect reaches far below the top predator alone; animals cannot make their own food. Because producers are the energy base every chain rests on, destroying them starves the whole web, not just the herbivores.

27. (C) Size is not the defining difference, a forest need not contain a particular predator, and forests are full of animals rather than free of them. What sets a forest apart is being a diverse, layered, self-renewing community lasting years, unlike a uniform crop sown and harvested in a single season.

28. (A) Floor plants face the least sun, not the harshest, so water storage for sun is misplaced; plants cannot make food without light by drinking humus; and the floor is shaded, not flooded with sun. Broad, thin leaves maximise capture of the scarce filtered light, letting the shade fern still photosynthesise enough to live.

29. (A) A forest does far more than cast shade or shift one gas, and even one forest measurably affects its surroundings, so the single-factor and 'nothing' options fail. Forests jointly govern temperature, moisture, rainfall, soil, water flow and carbon dioxide, so removing one disturbs all of these together.

30. (D) Coming first does not make an organism a producer, feeding on a dead leaf is decomposing rather than herbivory of living plants, and the fungus is the one being eaten,

not a flesh-eater. The fungus lives by breaking down dead matter, the defining work of a decomposer.

31. (C) Meat-eaters trace their food through the zebra back to the grass, so they are not spared, and the effect reaches beyond the lion alone; predators cannot make their own food. Since the grass is the producer underpinning the whole web, its loss starves every animal that depends on it directly or indirectly.

32. (B) Burning consumes oxygen rather than leaving it to rise, the released gas is mainly carbon dioxide not nitrogen, and carbon does not vanish but is freed as gas. Burning the tract both releases the trees' stored carbon as carbon dioxide and removes the living trees that had been absorbing it, so the gas climbs from both effects.

33. (B) Mice still eat seeds, so seeds are not left unconsumed; owl and fox each had two prey, so neither starves; and sparrows were not the mice's food. With fewer sparrows, both predators turn more to mice, pressing mouse numbers and shifting grazing pressure on the seeds.

34. (C) A forest is not a mere timber stock, nor a set of unrelated trees, nor a yearly human rebuild. Its many services arise precisely because it is one living, interlinked system in which plants, animals, soil and microbes work together, each part enabling the rest.

35. (C) Plants do not manufacture mineral nutrients from sunlight, the nutrients really are taken up into plant tissue rather than just borrowed in place, and rain does not resupply them daily. The soil stays stocked because decomposers continually recycle nutrients back from litter, dung and dead matter, balancing what plants remove.

36. (D) The filtered light is helpful, not harmful, it does feed plants and not merely warm animals, and it does not remove a plant's need for water and nutrients. Shade-adapted lower plants depend on exactly this weak filtered light to photosynthesise enough to live under the canopy.

37. (B) The snake still eats lizards, so it does not collapse; beetles ate clover rather than controlling grasshoppers; and lizards do not eat clover. Losing beetles pushes the lizard onto grasshoppers (pressing them down) while the snake, still supplied with lizards, feels only a mild effect.

38. (D) Trees do not pull clouds down, do not seed clouds with smoke, and do not absorb all solar heat. They moisten the air through transpiration and cool the ground with shade and cover, and these together favour rain and temper the local climate, both lost when the forest goes.

39. (A) Erosion is not soil 'floating off' for lack of tree-drinking, baked-hard soil is not what resists washing away, and rain temperature does not govern erosion. The forest holds

soil because roots bind it and litter slows the rain into the ground, protection the bare, churned slope simply lacks.

40. (B) The owl also ate warblers, so it does not starve; shrews grazed caterpillars, not leaves, so leaves are not stripped; and caterpillars were eaten by, not dependent on, shrews. Removing shrews relieves warbler eggs (warblers may rise) and caterpillars (one fewer predator), and pushes the owl toward warblers.

41. (C) A forest is not a single-species place, its living diversity is not merely a product of soil minerals, and its animals could not thrive on bare cleared land. The forest's layered, varied habitats house countless interdependent species, so protecting the forest protects them all.

42. (D) The leopard had three prey, so halving one does not starve it; deer did not control hares; and hares and birds did not eat deer. With fewer deer the leopard hunts hares and birds more (pressing them) and grass formerly grazed by deer is spared somewhat, a typical predator-switching ripple.

43. (D) Plants cannot create mineral nutrients from sunlight, the forest is described as needing little outside input so rain-import is not the basis, and daily sea imports by animals is invented. The forest sustains itself because decomposers keep recycling the same nutrients from dead matter back into the soil for reuse.

44. (B) Carbon taken from the air is not simply repacked into the air between trees, nor mainly carried off in streams, nor lodged in animal fur. Photosynthesis builds it into wood, leaves and, through litter, into soil humus, which is where a growing forest holds most of its captured carbon.

45. (A) A burnt patch is neither doomed to stay bare nor able to become a full forest overnight, and it is not permanently locked as grassland. Regrowth proceeds in stages over years, herbs to shrubs to trees, as soil and community rebuild, showing the forest's slow, dynamic recovery.

46. (B) Losing pollinating bees does affect seed-set, plants do not gain by keeping nectar (they need the pollination), and snakes gain no benefit from fewer bees. The bees' loss starves birds (decline), reaches snakes a step later, and cuts pollination so the plants set fewer seeds, a ripple running both up and down.

47. (D) Forests do not convert rain to oxygen, do not seal the soil (litter and roots open it), and do not feed a well by sending water to the sea. They recharge groundwater by letting rain percolate slowly into the ground; cleared land sheds rain too fast to recharge, so the water table and well fall.

48. (C) Extra dung does not instantly create forest, forest species clearly do affect one another, and deer do not hunt tigers. Without the tiger the deer multiply and strip the

young plants by overgrazing, hampering the forest's renewal, the classic many-step ripple from removing one keystone species.

49. (D) Least light does not force least life, the floor is genuinely the darkest layer, and floor organisms do not make their own light. The dead matter and moisture that collect on the floor provide abundant food and shelter for decomposers and ground dwellers, so life is dense there despite the dimness.

50. (C) The heron does not starve because the litter chain still feeds the fish; the fish is not wholly unaffected, since one of its two food routes is hit; and algae do not produce leaf litter. Killing the algae starves tadpoles (decline) but leaves the fish a second, litter-fed food source, so the upper levels are squeezed, not eliminated.

51. (A) It is not about a geometrically equal circle, nor a three-organism limit, and nutrients are reused rather than destroyed-and-remade. The nutrient loop is a cycle because the same nutrients keep returning to the soil for reuse, whereas the food chain runs one way with energy progressively lost.

52. (C) The paired contrast is not chance, the differences are not limited to flooding, and the cleared hill is plainly worse off. One cause, keeping versus losing the forest, jointly governs water flow, local climate and soil fertility, so the forested village gains all these services together while the cleared one loses them.

53. (C) Plastic and petrol are manufactured or fossil products, not gathered from a living forest, so any list containing them is wrong. Honey, gum, medicinal herbs and timber all come directly from forest plants and animals, making them the genuine forest products.

54. (A) Snakes did not protect grass, the eagle ate snakes rather than competing with them for frogs, and mid-chain removals do ripple outward. Killing snakes starves the eagle (decline) and releases frogs to rise, which then press grasshoppers down and ease the grazing pressure on the grass, ripples in both directions.

55. (D) Reducing the forest to saleable wood, to wildlife only, or to a mere stand of trees each ignores most of what it does. A forest is a lifeline because it delivers oxygen, nutrient recycling, soil and water protection, climate balance and shelter for many species, all together.

56. (C) A single tree's branchy top is its crown, not a 'forest crown,' and the meeting of many crowns overhead is the canopy, not the understorey or floor. Meera describes the canopy formed by many interlocking crowns; Vikram describes one tree's crown.

57. (B) Foxes did not keep rabbits off the grass nor feed the owl or the voles. Losing foxes frees rabbits (their only predator gone, so they rise) and removes a competitor for voles, leaving the owl more vole prey, a two-step ripple in two directions.

58. (D) A single-species stand, the mere presence of one predator, or a grass carpet each miss the essence of a forest. The decisive evidence is a diverse, layered, self-renewing community of plants, animals and microbes persisting over the years, which is what makes a place a forest.

59. (A) A tiger is a carnivore not a producer, grass is a producer not a carnivore, and a fungus is a decomposer not a herbivore. Only the mushroom-as-decomposer pairing is right, since the fungus breaks dead matter into humus.

60. (B) Removing predators does not improve the forest, decomposers are essential, and the two unrelated losses do not cancel. Losing the predator lets herbivores overgraze the plants while losing decomposers stalls nutrient recycling and starves the soil, so the forest suffers on both fronts together.